UNIVERSITY OF TORONTO
Faculty of Arts & Science

FALL 2023 EXAMINATIONS

CSC 311 H1F

Duration: 3 hours

Aids Allowed: One *double*-sided *8.5"×11"* reference sheet and one non-programmable calculator.

*Do **not** turn this page until you have received the signal to start.*
In the meantime, write your name, and student number below (please do this now!) and *carefully* read *all* the information on the rest of this page.

First name (please write as legibly as possible within the boxes)

Last name

Student ID number

- Turn off and place all cell phones, smart watches, electronic devices, and unauthorized study materials in your bag under your desk. If it is left in your pocket, it may be an academic offence.
- You must obtain a grade of at least 40% on the final exam to pass the course.
- When you are done with your exam, raise your hand for someone to come and collect your exam. *Do not collect your bag and jacket before your exam is handed in.*
- If you are feeling ill and unable to finish your exam, please bring it to the attention of an Exam Facilitator so it can be recorded before leaving the exam hall.
- In the event of a fire alarm, do **not** check your cell phone when escorted outside.
- This final examination consists of 5 questions worth 80 marks on 18 pages (including this one), printed on both sides of the paper. *When you receive the signal to start, please make sure that your copy of the examination is complete.*
- Answer each question directly on the examination paper, in the space provided, and use a "blank" page for rough work. If you need more space for one of your solutions, use one of the "blank" pages and *indicate clearly the part of your work that should be marked.*
- As a student, you help create a fair and inclusive writing environment. If you possess an unauthorized aid during an exam, you may be charged with an academic offence.

STUDENTS MUST HAND IN ALL EXAMINATION MATERIALS AT THE END

Question 1. Short Answers [34 MARKS]

This question has *many* ((a),. . . ,(p)) subparts – please check that you have gone through and completed all of them prior to handing in your final exam.

Part (a) [4 MARKS]

A good strategy for model selection is often to choose the hyperparameter that minimizes the cost function on a validation set. For each of the following models, hyperparameters, and cost functions, say whether it's a good idea to choose the hyperparameter by minimizing the cost on a validation set.

Your answer should take the form of <1/2> - <Yes/No>: <one/two line explanation>

1. Model: K-means. Hyperparameter: number of clusters. Cost function: sum of squared distances to the nearest cluster center (distortion).
2. Model: PCA. Hyperparameter: number of principal components. Cost function: reconstruction error.

Part (b) [2 MARKS]

Suppose you are trying to learn the parameters of the conditional probability of a word given a class within a Naive Bayes model. Bayesian parameter estimation using a uniform prior over the weights of the model will make the same predictions (in terms of the probability assigned to test examples) as maximum likelihood estimation.

Circle the correct answer: True or False

Explain briefly:

Part (c) [2 marks]

Naive Bayes model (as covered in lecture) is a parametric model.

Circle the correct answer: True or False

Explain briefly:

Part (d) [1 mark]

Finite differences is useful for testing the implementation of a gradient computation. Give one reason why we don't use finite differences to compute all our gradients when we run gradient descent.

Part (e) [1 mark]

State one difference between reinforcement learning and supervised learning.

Part (f) [2 marks]

There is a unique set of optimal weights for any linear regression problem (You can assume that the data matrix $X \in \mathbb{R}^{N \times D}$ has rank D).

Circle the correct answer: True or False

Explain briefly:

Part (g) [2 MARKS]

There is a unique set of optimal weights for any feed-forward neural network.

Circle the correct answer: True or False

Explain briefly:

Part (h) [4 MARKS]

Consider the following probabilistic model where the data is generated with the procedure below.
The dataset has N data points $\{(x^1, t^1), (x^2, t^2), ..., (x^N, t^N)\}$, where x^i is the input and t^i is a binary target.

$$t|x \sim B(\sigma(w^\top x)),$$

where $B(\theta)$ is a Bernoulli distribution and σ is the logistic sigmoid function.

1. Write down the log likelihood $p(t|x)$ for the whole dataset.

   Your answer should take the form $\sum_{i=1}^{N} \kappa(x_i, w)$ where κ is a formula involving the inputs and weights.

2. If we do maximum likelihood estimation, which model discussed in lecture does the above correspond to?

Part (i) [2 marks]

Suppose that we want to group N data points $\{x^1, x^2, \ldots, x^N\}$ into K clusters. K-means is guaranteed to find the optimal cluster centers m_k and cluster assignments r_k^i that minimize $\sum_{i=1}^{N}\sum_{k=1}^{K} r_k^i ||x^i - m_k||^2$.

Circle the correct answer: True or False

Explain briefly:

Part (j) [2 marks]

(True or False) K-means is guaranteed to terminate (converge).

Circle the correct answer: True or False

Explain:

Part (k) [2 marks]

(True or False) There is a setting for the β parameter in soft K-means so that the solution from soft K-means is the same as hard K-means.

Circle the correct answer: True or False

Explain:

Part (l) [1 MARK]

Provide one reason why we could not have used PCA to solve the Netflix Prize recommender system challenge competition.

Part (m) [2 MARKS]

We want train a logistic regression classifier to predict $y \in \{Y, N\}$ in the following dataset. Is it a good idea to apply PCA to the dataset before training the logistic regression classifier?

x_1	x_2	y
10	1	Y
-10	1	Y
10	-1	N
-10	-1	N

Table 1: 2D data set for training a logistic regression classifier

Circle the correct answer: It (IS or ISN'T) a good idea to apply PCA before logistic regression.

Explain briefly:

Part (n) [2 marks]

In a k-nearest neighbours classifier, what happens to the bias and variance as $k \to \infty$?

Bias: (Increases / Decreases)

Explain briefly:

Variance: (Increases / Decreases)

Explain briefly:

Part (o) [2 marks]

When doing maximum likelihood estimation, why do we often maximize the *log* likelihood, rather than the likelihood function directly?

Explain briefly:

Part (p) [2 marks]

You are trying to solve a (balanced) binary classification problem with inputs x and targets $t \in \{0, 1\}$. The data generating process is: $x \sim \mathcal{N}(0, 1)$ if $t = 0; x \sim \mathcal{N}(3, 1)$ if $t = 1$. What is the Bayes optimal prediction for a new query point x assuming 0-1 loss? Your answer should take the form of $y = f(x)$ where the function f is described mathematically. Provide a one-line justification for your answer.

Part (q) [1 mark]

State one difference between the results obtained from clustering using soft-Kmeans and hard Kmeans.

x_1	x_2	y
0	0	1
0	1	0
1	0	0
1	1	1

Table 2: XNOR dataset

Question 2. Binary Linear Classification [12 MARKS]

Consider the following dataset representing the XNOR boolean function.

Part (a) [1 MARK]

Is the data linearly separable?

Part (b) [2 MARKS]

Briefly justify your choice on whether the data is (or is not) linearly separable.

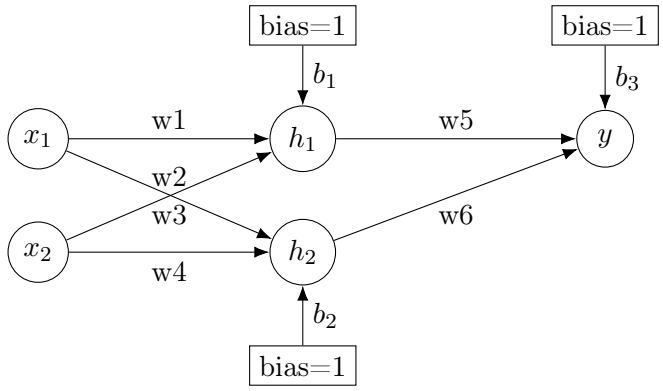


Figure 1: Two layer neural network with threshold activation $\phi(x) = \mathbb{I}[x > 0]$

Part (c) [9 marks]

Design a two layer multi-layer perceptron using the hard threshold activation function $\phi(x) = \mathbb{I}[x > 0]$ to perfectly classify the above dataset in Table 2 (XNOR dataset).

Then fill in the weights in the table below.

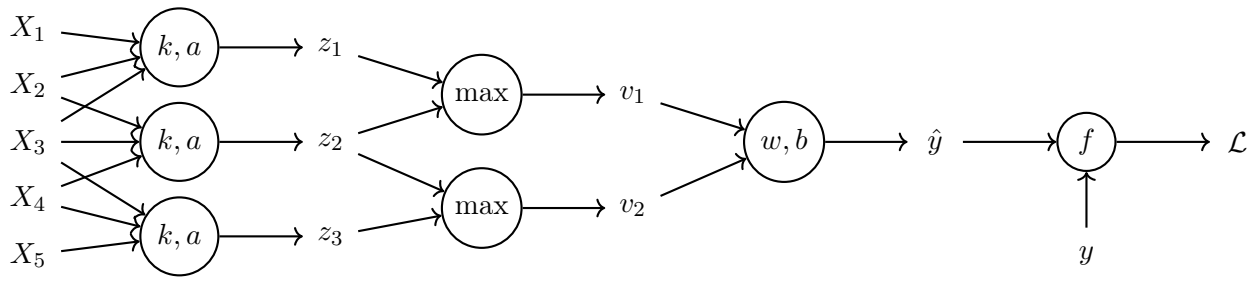
While no justification is required to receive full marks, explaining what you intend the hidden units to do may help you receive partial credit.

Parameter	**Value**
w1	
w2	
w3	
w4	
w5	
w6	
b1	
b2	
b3	

Table 3: Neural Network Parameters

Question 3. Convolutional Neural Networks and Backpropagation [12 MARKS]

The following is the computation graph for a simple convolutional neural network. [1]



Nodes with a circle denote operations; nodes without a circle denote variables that are computed during the forward pass. The network performs the following operations left-to-right:

- **Convolutional Layer**: the convolution layer performs the following computation:

$$\begin{bmatrix} z_1 \\ z_2 \\ z_3 \end{bmatrix} = \begin{bmatrix} k_1 & k_2 & k_3 & 0 & 0 \\ 0 & k_1 & k_2 & k_3 & 0 \\ 0 & 0 & k_1 & k_2 & k_3 \end{bmatrix} \begin{bmatrix} X_1 \\ X_2 \\ X_3 \\ X_4 \\ X_5 \end{bmatrix} + \begin{bmatrix} a \\ a \\ a \end{bmatrix} \tag{1}$$

- **Max-Pooling Layer**: the max-pooling layer performs a max-pooling and ReLU operation, as follows:

$$\begin{bmatrix} v_1 \\ v_2 \end{bmatrix} = \begin{bmatrix} \max(z_1, z_2, 0) \\ \max(z_2, z_3, 0) \end{bmatrix} \tag{2}$$

- **Fully-Connected Layer**: the fully-connected layer transforms v_1, v_2 into $\hat{y}$ as follows:

$$\hat{y} = \begin{bmatrix} w_1 & w_2 \end{bmatrix} \begin{bmatrix} v_1 \\ v_2 \end{bmatrix} + b \tag{3}$$

- **Loss Function**: finally, the network uses the squared loss function. That is,

$$\mathcal{L} = \frac{1}{2}(y - \hat{y})^2 \tag{4}$$

Part (a) How many parameters does this model contain? [1 MARK]

[1] This question is inspired by Stanford's CS231N.

Part (b) Use bar notation (e.g., $\overline{w_1} = \frac{\partial \mathcal{L}}{\partial w_1}$) to write out the following gradients: [4 MARKS]

$\overline{w_1} =$ (5)

$\overline{w_2} =$ (6)

$\overline{b} =$ (7)

Part (c) Given $\overline{v_1}, \overline{v_2}$, write out the following gradients: [3 MARKS]

$\overline{z_1} =$ (8)

$\overline{z_2} =$ (9)

$\overline{z_3} =$ (10)

(11)

Part (d) Finally, given $\overline{z_1}, \overline{z_2}, \overline{z_3}$, write out the following gradients: [4 MARKS]

$\overline{k_1} =$ (12)

$\overline{k_2} =$ (13)

$\overline{k_3} =$ (14)

$\overline{a} =$ (15)

Question 4. Principal Component Analysis [11 MARKS]

We are interested in using PCA to reduce the number of features of a 3-dimensional dataset.

The empirical mean of the data is $\hat{\mu} = \begin{pmatrix} 1 & 0 & 1 \end{pmatrix}$, and the empirical covariance matrix has the following spectral decomposition:

$$\hat{\Sigma} = \begin{bmatrix} \frac{1}{2} & \frac{1}{2} & 0 \\ 0 & 0 & 1 \\ \frac{1}{2} & -\frac{1}{2} & 0 \end{bmatrix} \begin{bmatrix} 5 & 0 & 0 \\ 0 & 0.1 & 0 \\ 0 & 0 & 0.01 \end{bmatrix} \begin{bmatrix} \frac{1}{2} & \frac{1}{2} & 0 \\ 0 & 0 & 1 \\ \frac{1}{2} & -\frac{1}{2} & 0 \end{bmatrix}^\top$$

Part (a) [3 MARKS]

What is the first (top) principal component? Why? Give your answer as a vector.

Part (b) [2 MARKS]

In lecture, we discussed that data often lie on a low-dimensional space. What is the dimensionality of the space that this dataset lies on? Why?

Part (c) [6 MARKS]

Consider the vector $x = \begin{pmatrix} 1 & -1 & 2 \end{pmatrix}$. Determine the reconstruction $\tilde{x}$ after projecting x onto the principal subspace spanned by the solution to PCA (with $K = 1$). Your answer should be a vector of fractions. Show all your work to receive full credit.

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Question 5. Gaussian Discrimination Analysis [11 MARKS]

In this question we will handle the problem of binary classification using Gaussian discriminant analysis.

Part (a) [3 MARKS]

Answer the following questions – one or two lines will suffice in response to each.

1. List one assumption made by Gaussian Discriminant Analysis.
2. How does Gaussian Discriminant Analysis differ from Logistic Regression?
3. List one setting where you expect logistic regression to work better than GDA?

Part (b) [2 MARKS]

Recall that the multivariate Gaussian pdf is given by $p(x; \mu, \Sigma) = \frac{1}{\sqrt{(2\pi)^D |\Sigma|}} \exp(\frac{-1}{2}(x-\mu)^T \Sigma^{-1}(x-\mu))$.

1. Derive an expression for $\log p(y=1|x)$ that GDA uses to calculate the probability of the class label in terms of $\mu_1, \Sigma_1, p(y=1)$.
2. What rule of probabilities do you use to derive this expression?

Your final answer should be left in the form: $\log p(y=1|x) = g(y=1, x) - \log p(x)$.

Part (c) [6 MARKS]

Consider a dataset with two features and two classes. The prior probabilities of the labels are $P(y = 0) = 0.4$ and $P(y = 1) = 0.6$. The mean and precision matrices of each class are given as follows:

- Class 0: $\mu_0 = [2, 4]$, $\Sigma_0^{-1} = \begin{bmatrix} 1 & 0.5 \\ 0.5 & 2 \end{bmatrix}$
- Class 1: $\mu_1 = [3, 3]$, $\Sigma_1^{-1} = \begin{bmatrix} 1 & 0.5 \\ 0.5 & 2 \end{bmatrix}$

What class would GDA predict at the point $x = [2, 2]$? You are required to show *all* your work and calculations to receive full marks.

Page to continue working.

End of Exam